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Introduction

MDB3 | MTD.ba VZ WS26

Hagenberg 01./02.10.2026

HAGENBERG | LINZ | STEYR | WELS



UNIVERSITY
OF APPLIED SCIENCES
UPPER AUSTRIA

Course Instructor

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- Lectures since 2015
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What We Cover in Moderne Datenbanken

1. **SQL hurts on flexible data:** *Data that Hurts in SQL, Document Modelling*
2. **Documents:** *CRUD and Queries, Aggregation I, Aggregation II*
3. **The price of documents:** *Indexes and Schema Validation, Transactions and Consistency*
4. **SQL back, at scale:** *NewSQL I: Concepts and Systems, NewSQL II: A Cluster in Your Hands*
5. **A problem neither solves well:** *Search I: How Search Works, Search II: Analysis and Aggregations*
6. **Which one for what:** *Polyglot Persistence*

Three systems, one dataset

MongoDB, CockroachDB and Elasticsearch, all from PHP, all on the same catalogue of 4,456 items and 391 villagers. You know SQL from *Relational Databases*; every new idea starts from what SQL does and what it cannot.

Eight Sessions

	Gr. 1	Gr. 2	Session
S1	Thu 1 Oct	Fri 2 Oct	Data that Hurts in SQL; Document Modelling
S2	Mon 5 Oct	Wed 7 Oct	CRUD and Queries; Aggregation I
S3	Tue 13 Oct	Tue 13 Oct	Aggregation II; Indexes and Schema Validation
S4	Thu 22 Oct	Wed 21 Oct	Transactions and Consistency; NewSQL I
S5	Thu 19 Nov	Thu 19 Nov	NewSQL II: a cluster in your hands
S6	Mon 23 Nov	Mon 23 Nov	Search I: how search works
S7	Tue 24 Nov	Tue 24 Nov	Search II: analysis and aggregations
S8	Tue 15 Dec	Tue 15 Dec	Polyglot persistence; exam Q&A; open lab

Organisational Matters

Attendance and participation

- This is an integrated course (ILV): **attendance is mandatory** and is recorded in every session
- Below **80 % attendance** you have to complete a **compensatory exercise** to make up the missed hours
- Participation is expected; it can only influence your grade positively

Material

- Slides, the stack and the dataset are on Moodle
- Two groups, the same programme in both; you may visit the other group's session if you miss yours

Grading

Homework: 50 %

- Four sheets of **24 points** (96 in total)
- A01 MongoDB: modelling and CRUD, due 20.10.2026
- A02 MongoDB: aggregation and validation, due 09.11.2026
- A03 CockroachDB, due 06.12.2026
- A04 Elasticsearch, due 21.12.2026

Exam: 50 %

- Moodle, 60 minutes, **100 points**
- Six questions of 10 to 20 points, mostly free text: model this, read a pipeline, explain a guarantee, choose a system
- Date: tba

Rules

A positive grade needs at least 50 % in each half. Homework is handed in on Moodle; late means minus 4 points per day. **AI tools are allowed and encouraged in the homework and you state what you used**; they are not allowed in the exam.

If You Fail

Attendance below 80 % or homework not passed

- You **must** submit **compensatory work**, and it must reach a passing grade

Exam failed or missed

- **Second attempt:** a second exam date
- Excused absence (sickness, other valid reasons): you take the exam on the second date without penalty

Second failure

- The third attempt is a **commissional examination**. It is only possible if the compensatory work has been completed

Effort and Tools

The effort

- 2 ECTS = 50 hours: 18 in the sessions, about 24 for the four sheets, about 8 to prepare the exam
- Please note your hours; it lets me size the sheets for next year

Your toolbox

- Docker with the course stack: MongoDB, CockroachDB, Elasticsearch and Kibana, PHP 8.4 (**mdb3-stack.zip** on Moodle, installed **before** the first session)
- MongoDB Compass, the free desktop client
- Your editor of choice; **mongosh** inside the stack for the database shell

Why It Matters

- Almost no application runs on one database any more: a relational core, a document store for the flexible parts, a search engine for the search box, a cache in front
- The job is not “learn MongoDB”; it is knowing **which shape of problem each system is built for** and **what its guarantees cost**
- After this course you can model data as documents, write aggregation pipelines, run and break a distributed SQL cluster, build a ranked search, and argue for an architecture